MATH317 · NUMERICAL METHODS

Mock Midterm

A practice exam in the midterm's format · 60 minutes · 30 marks

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01 Before you start

This exam is original. It follows the shape of the four past midterms, with new questions and new numbers.

- Set a 60-minute timer.
- Solve on paper and write every step.
- Put the calculator in radian mode.
- The solutions are in a separate file. Open it only after you finish.

02 Questions

Q1 TRUE / FALSE · 7 M

1) The relative error is the absolute error divided by the approximate value.
2) Chopping to n digits discards all digits after the n-th digit.
3) A Maclaurin series is a Taylor series expanded about c = 1.
4) In single precision the bias of the exponent is 127.
5) Machine epsilon is the largest number that a computer can store.
6) In the bisection method the error bound is halved at every step.
7) Newton's method needs both f(x) and f′(x) at every step.

Q2 HORNER + TAYLOR · 3 + 5 M

Given $P(x) = 2x^3 - 5x^2 + 7$.

a) Use Horner's algorithm to evaluate P(2).

b) Using the complete Horner's algorithm, find the Taylor expansion of P(x) about r = 1.

Q3 FLOATING-POINT · 7 M

Determine the single-precision machine representation in a 32-bit word-length computer of the decimal number −37.40625.

Q4 BISECTION · 2 + 6 M

Given $f(x) = x^3 + x - 3$.

a) Prove that f has exactly one real root in the interval [1, 2].

b) Find a root on [1, 2] using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than 0.5×10^{-6}.

Q5 NEWTON · BONUS

For the same $f(x) = x^3 + x - 3$, start from $x_0 = 1$ and apply two iterations of Newton's method.